Avoiding sillies

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A motivating quote

To succeed in competition math it is not enough to be right. It is also necessary to not be wrong.

What is a silly?

Definition

A *silly* is an almost-correct solution with a "dumb" mistake.

Sillies are Minnetonka's downfall.

Examples

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- Arithmetic mistakes
- Forgetting edge cases
- Misreading the question
- Sign errors

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A *silly* is an almost-correct solution with a "dumb" mistake.

- Arithmetic mistakes
- Forgetting edge cases
- Misreading the question
- Sign errors
- Listening to a song with a number and writing that instead

Non-examples

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Significant mistakes are not sillies. If there are n problems and you scored m, you didn't silly $n - m$ problems.

- Fundamentally flawed approach
- Failed guess

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- Fundamentally flawed approach
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- Answer sheet spontaneously catches fire
- Accidentally eat chocolate containing alcohol before the contest

The problem

Sillies are easily correctable.

Corollary

Reducing sillies is an easy way to improve.

Example: Me at HMMT 2025

Factors increasing sillies

- Doing problems quickly (Me at ARML tiebreaks 2025)
- Headsolving or solving with low scratch paper
- Sleep deprivation or jet lag
- Strong, distracting emotions, such as excitement, anger, or heartbreak
- Sickness

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- Having family history of sillying

Factors ~~increasing~~ decreasing sillies

- Doing problems ~~quickly~~ slowly
- ~~Headsolving or solving with low scratch paper~~ using scratch paper liberally
- ~~Strong, distracting emotions, such as excitement, anger, or heartbreak~~ being locked in

Most important advice

Reasoning must be PAINFULLY explicit. Smart people make stupid mistakes. Write down all reasoning. Do not skip steps in your head!

Slow is smooth, smooth is fast. Sillies when barying ISL 2024 G4 turned 20 minutes into 3 hours.

Pointing-and-calling

- In many Asian railways, conductors are required to literally point and call out what they're seeing and doing
- Sounds dumb? Shown to reduce error by 85%
- Being painfully explicit reduces errors

The pointing-and-calling method



Figure: The process of pointing and calling. Image from Wikipedia.

Good and bad scratchwork

Scratchwork isn't only for computation. Work out small examples. Write down reasoning.

Writing everything down is pointing-and-calling. If you think 2×3, don't write 6. Don't skip steps!

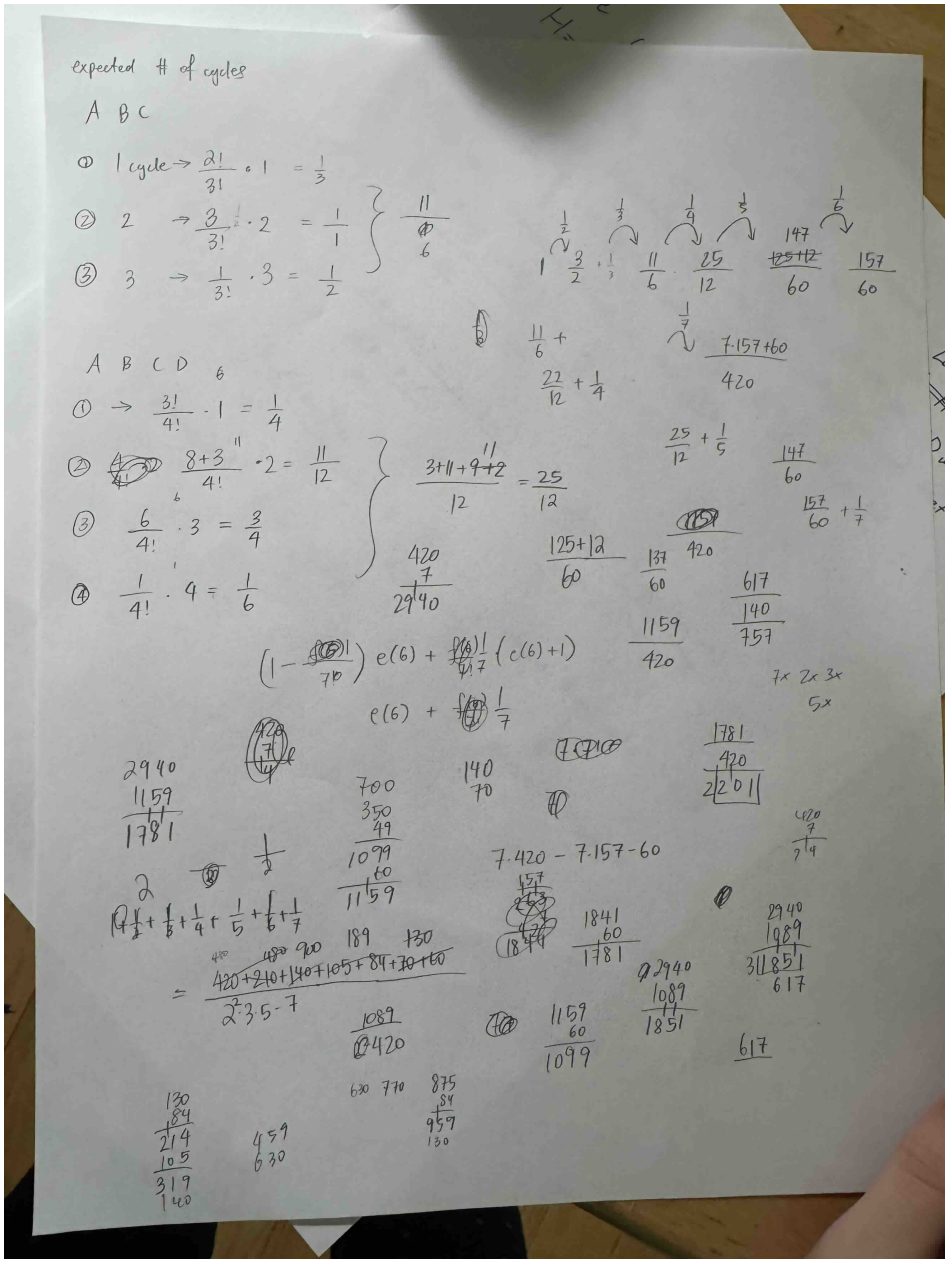
Don't use test paper margins (not enough space). You'll write over yourself and problems. You are not Fermat.

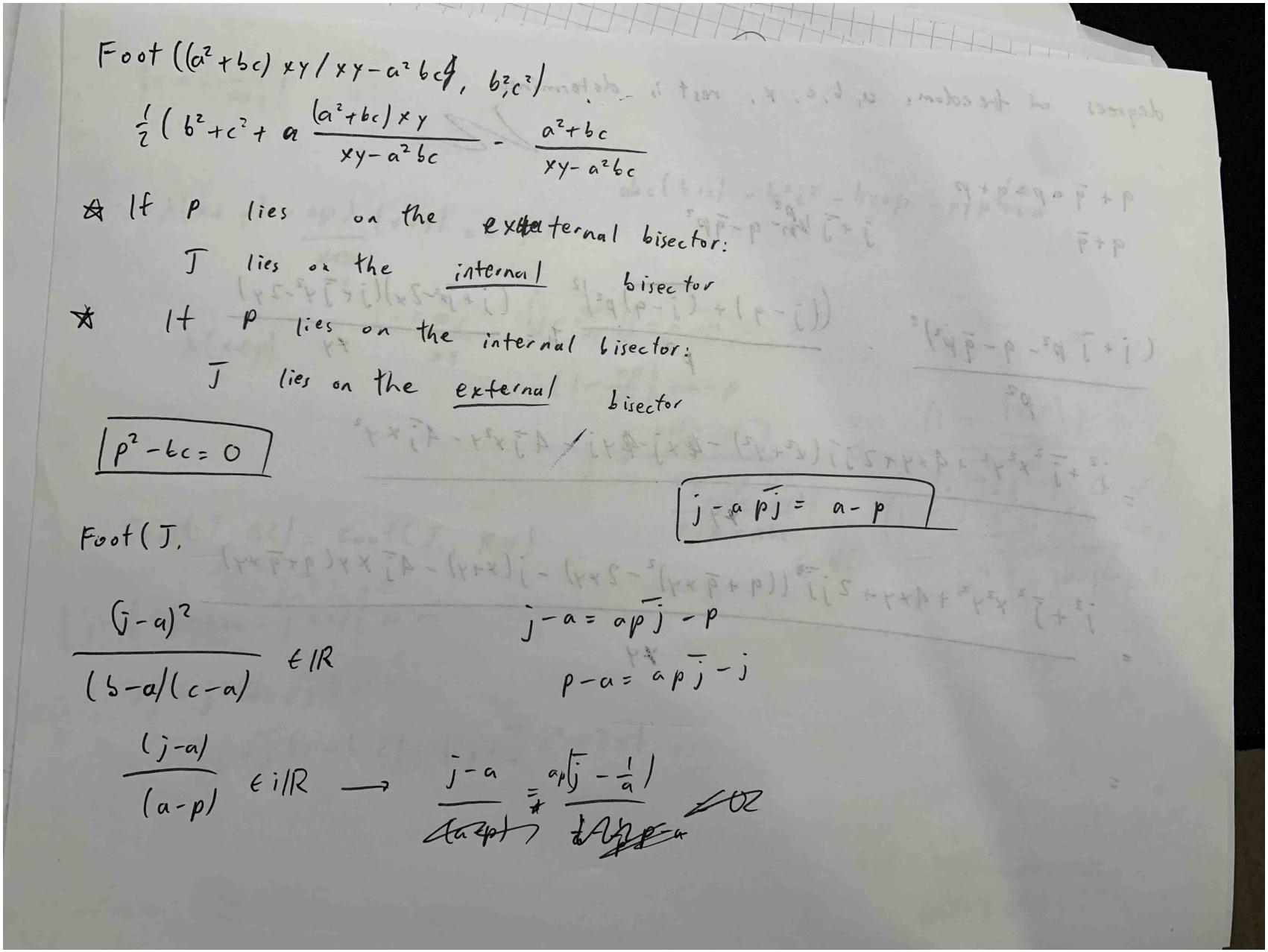
Good and bad scratchwork

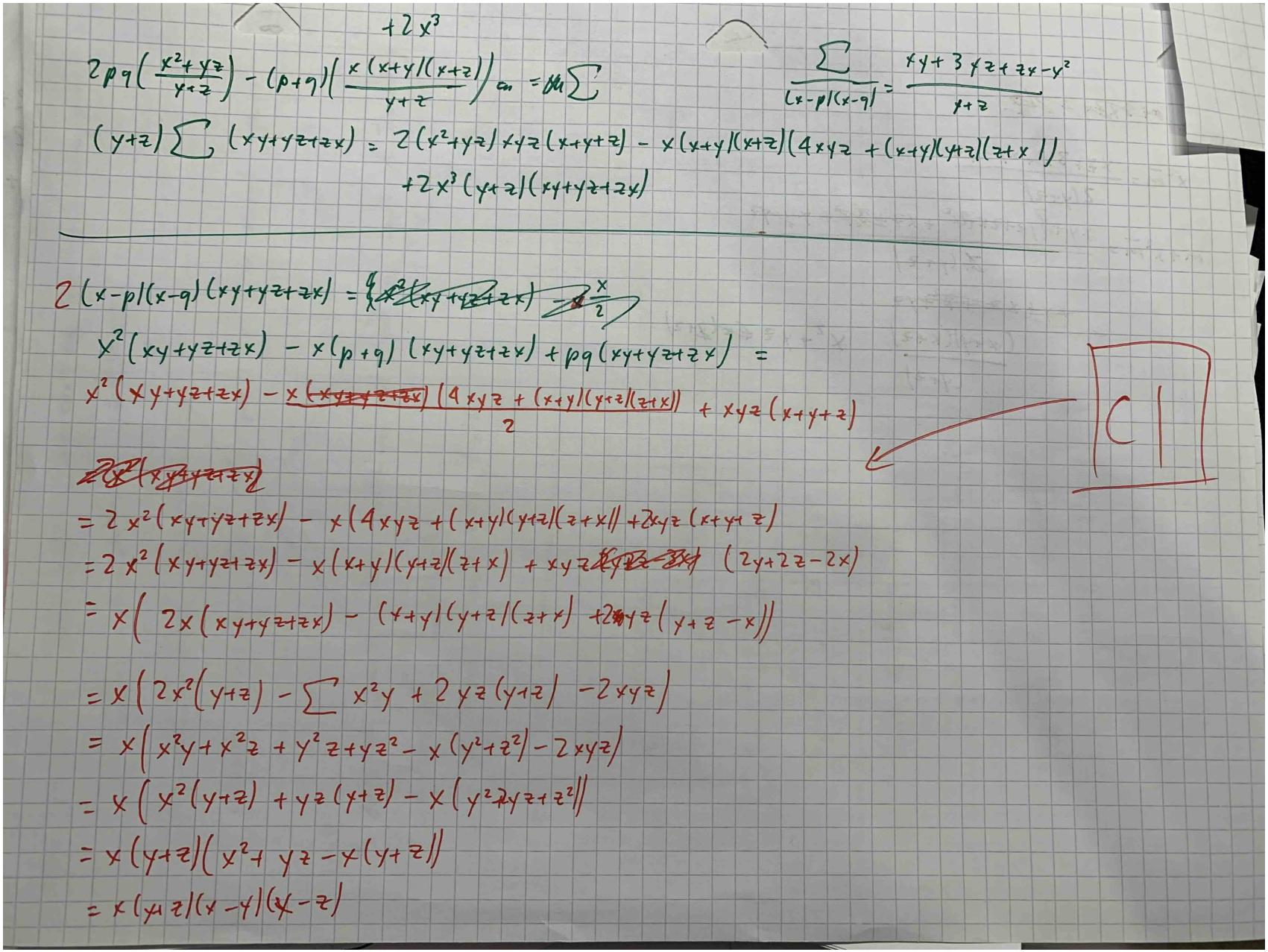
The following slides include examples of scratch paper.

Think about how their paper is organized and how work is being done.

Lanie Deng




Foot((a²+bc) xy / xy − a²bc, b², c²)
½ (b² + c² + a (a²+bc) xy / (xy − a²bc) − (a²+bc) / (xy − a²bc))
★ If P lies on the external bisector:
J lies on the internal bisector
★ If P lies on the internal bisector:
J lies on the external bisector
p² − bc = 0
j − a p̄ j = a − p
Foot(J,
(j − a)² / (b − a)(c − a) ∈ ℝ
j − a = a p̄ j − p
p − a = a p̄ j − j
(j − a) / (a − p) ∈ iℝ ⟶ j − a / a p (j̄ − 1/a)



$$2pq\left(\frac{x^2+yz}{y+z}\right) - (p+q)\left(\frac{x(x+y)(x+z)}{y+z}\right)_{cyc} = \sum \frac{+2x^3}{(x-p)(x-q)}$$

$$\sum \frac{xy+3yz+zx-y^2}{y+z}$$

$$(y+z)\sum (xy+yz+zx) = 2(x^2+yz)(xyz(x+y+z) - x(x+y)(x+z)(4xyz + (x+y)(y+z)(z+x)) + 2x^3(y+z)(xy+yz+zx)$$

$$2(x-p)(x-q)(xy+yz+zx) = \cancel{x^2(xy+yz+zx)} - \frac{x}{2}$$

$$x^2(xy+yz+zx) - x(p+q)(xy+yz+zx) + pq(xy+yz+zx) =$$

$$x^2(xy+yz+zx) - \frac{x(xy+yz+zx)(4xyz + (x+y)(y+z)(z+x))}{2} + xyz(x+y+z)$$

$$= 2x^2(xy+yz+zx) - x(4xyz + (x+y)(y+z)(z+x)) + 2xyz(x+y+z)$$

$$= 2x^2(xy+yz+zx) - x((x+y)(y+z)(z+x) + xyz(2y+2z-2x)$$

$$= x(2x(xy+yz+zx) - ((x+y)(y+z)(z+x) + 2yz(y+z-x))$$

$$= x(2x^2(y+z) - \sum x^2y + 2yz(y+z) - 2xyz)$$

$$= x(x^2y+x^2z+y^2z+yz^2 - x(y^2+z^2) - 2xyz)$$

$$= x(x^2(y+z) + yz(y+z) - x(y^2+2yz+z^2))$$

$$= x(y+z)(x^2+yz-x(y+z))$$

$$= x(y+z)(x-y)(x-z)$$

- if $2xy$ is a perfect square!

$x = \frac{99^2}{2}$: 20, 21, 22, 23 work

$x + \log_8(y) = 1024 \rightarrow$

$y = 8^{1024-x} \sim 2^{1024-x}$

- if x is even, then $2xy = \sim 2x \cdot$ ~~2^{1024-x}~~ is a square

- if x is odd, then $2xy \sim 2x 2^{1024-x} \sim x$ is a square

$1^2, 3^2, 5^2, \ldots,$ ~~31^2~~ < 1024 $\quad 16 \times 22 = 38$

$\frac{2^2}{2}, \frac{4^2}{2}, \ldots, \frac{44^2}{2}$ $\quad \left(\frac{45^2}{2}\right) > 1024$

$46^2 > 2048$

N, $N+2019$ are palindromes

The last digit goes down by 1

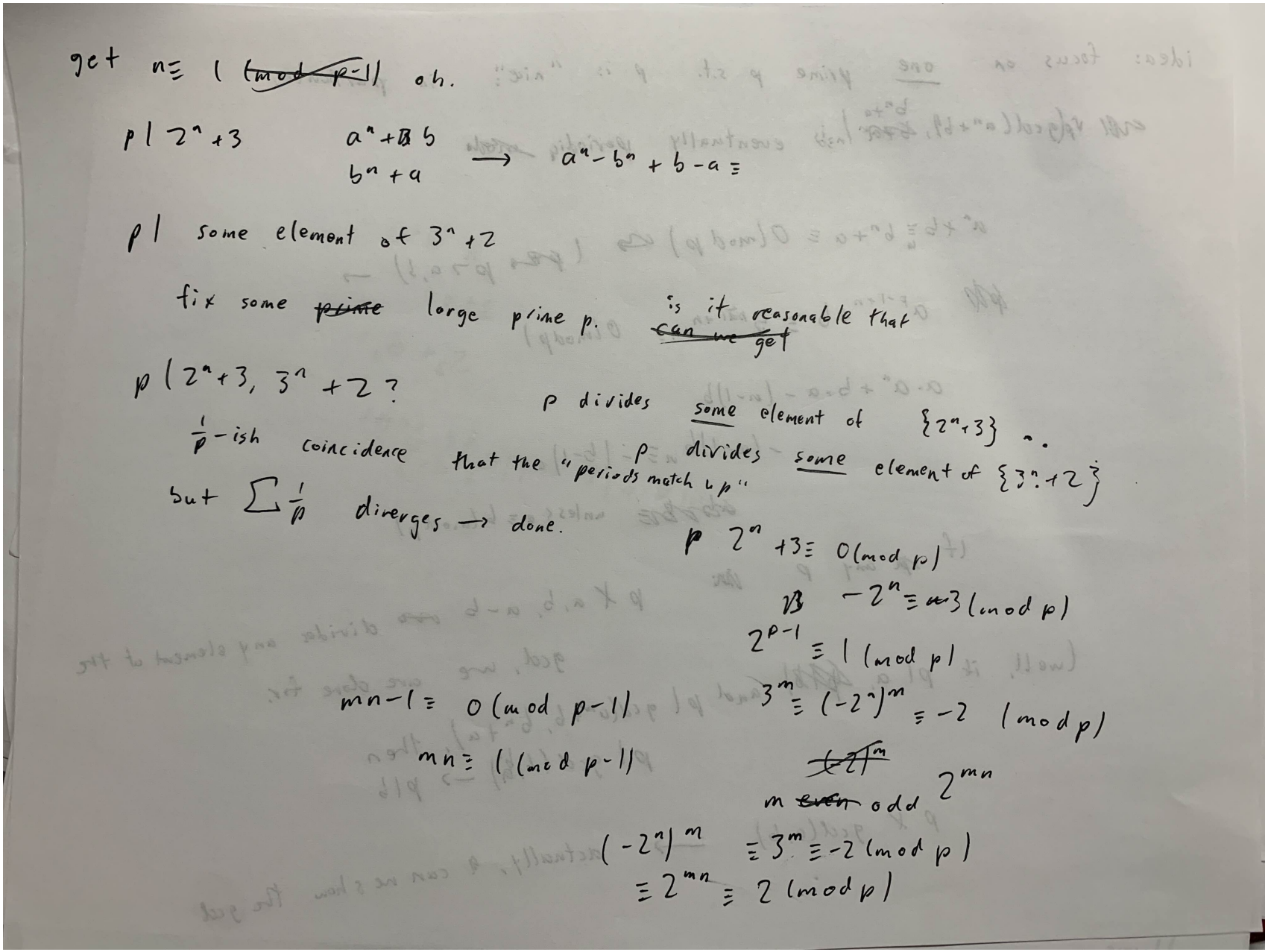
$$\begin{array}{r} \underline{a\ b\ c} \quad \underline{c\ b\ a} \\ 2 \quad 0\ 1\ 9 \end{array}$$

2019

(last digit can't be 0) $\quad 1 \rightarrow 0$, etc.

hence there needs to be an "overfill"

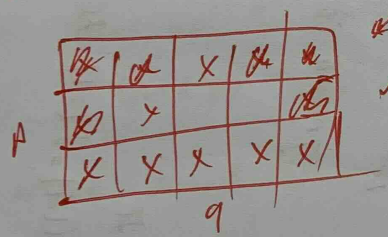
(in particular, we want $2 \rightarrow 1$)

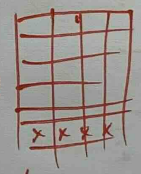

get $n \equiv 1 \pmod{p-1}$ oh.
$p \mid 2^n+3$ a^n+b^n b^n+a $\longrightarrow$ $a^n-b^n+b-a \equiv$
$p \mid$ some element of 3^n+2
fix some ~~prime~~ large prime p. is it reasonable that ~~can we get~~
$p \mid 2^n+3,\ 3^n+2$?
p divides some element of $\{2^n+3\}$ ..
$\frac{1}{p}$-ish coincidence that the "periods match up" p divides some element of $\{3^n+2\}$
but $\sum \frac{1}{p}$ diverges $\longrightarrow$ done.
$p \mid 2^n+3 \equiv 0 \pmod p$
$-2^n \equiv 3 \pmod p$
$2^{p-1} \equiv 1 \pmod p$
$3^m \equiv (-2^n)^m \equiv -2 \pmod p$
$mn-1 \equiv 0 \pmod{p-1}$
$mn \equiv 1 \pmod{p-1}$
~~$(-2)^m$~~ 2^{mn}
m ~~even~~ odd
$(-2^n)^m \equiv 3^m \equiv -2 \pmod p$
$\equiv 2^{mn} \equiv 2 \pmod p$

- The set must be surjective modulo all but at most one prime.
  (Proof. If else, then pull tube that cross-section, it's ambiguous)
- The graph must be connected.

Answer: $2 \cdot 89 - 1$.

Construction: Pick everything to be 0 mod 97. Then just make the grid mod pq for all else. look like this.

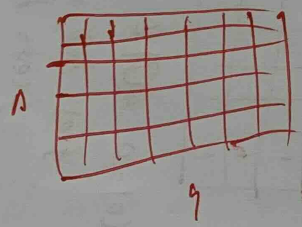




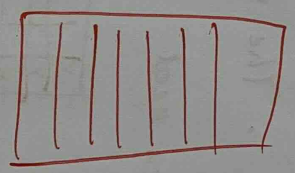
$p+q-1$

This works— it detects periodicity mod q always.

Bound. 87, 89, and 97, pick one,



$2 \cdot 87 - 1$ is the bound



$89+87-1 = 175$

$88 \cdot 2 = 16 \cdot 10 = 175.$

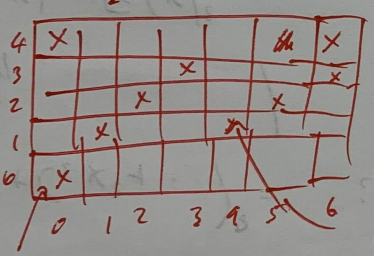
$p_1, \dots, p_n$ are the primes < 100.

we need to find (for all but 1)

$i \neq j$ s.t. $f(i) \neq f(j)$ ~~mod p~~ in $\mathbb{Z}/(q)$

$2 \cdot 89$ mod works, yet? The bound is not good — we can make it ambiguous between 89 and 97. Is the answer smaller?

Can we choose $2 \cdot 89 - 2$? (mod 5)



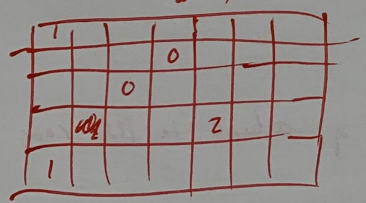
like, choose $q \mid 10$ such that

(mod 7) on its own

oh. we are playing "perfectly"

If we only pick q, what goes wrong?

Bob: ~~Forces~~ ~~f(6)~~ ~~Oops~~



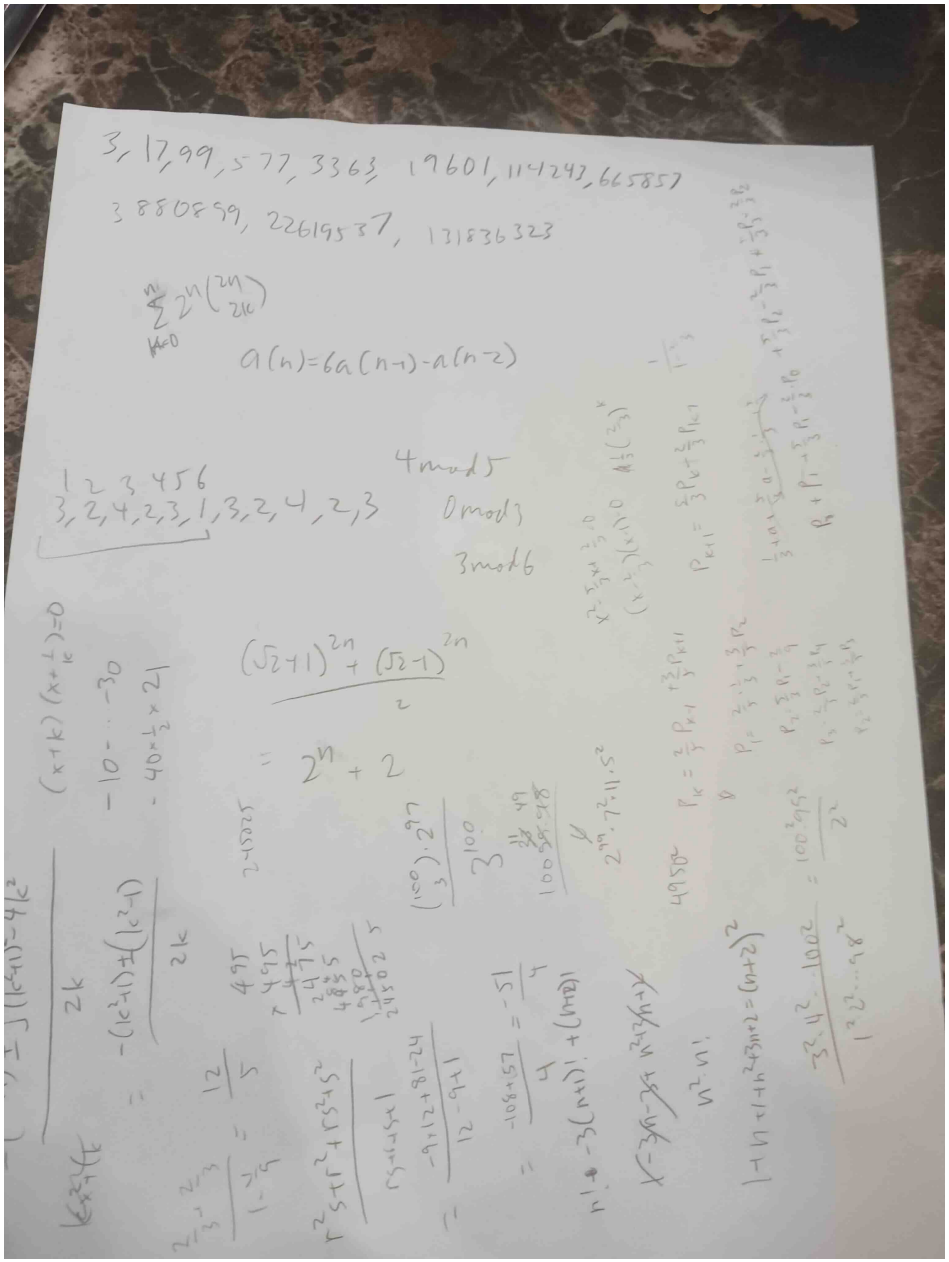
element on its own

select $2q-1$ squares such that it is ~~impossible~~ to tell them ~~apart~~ s.t. ambiguity if two elements are

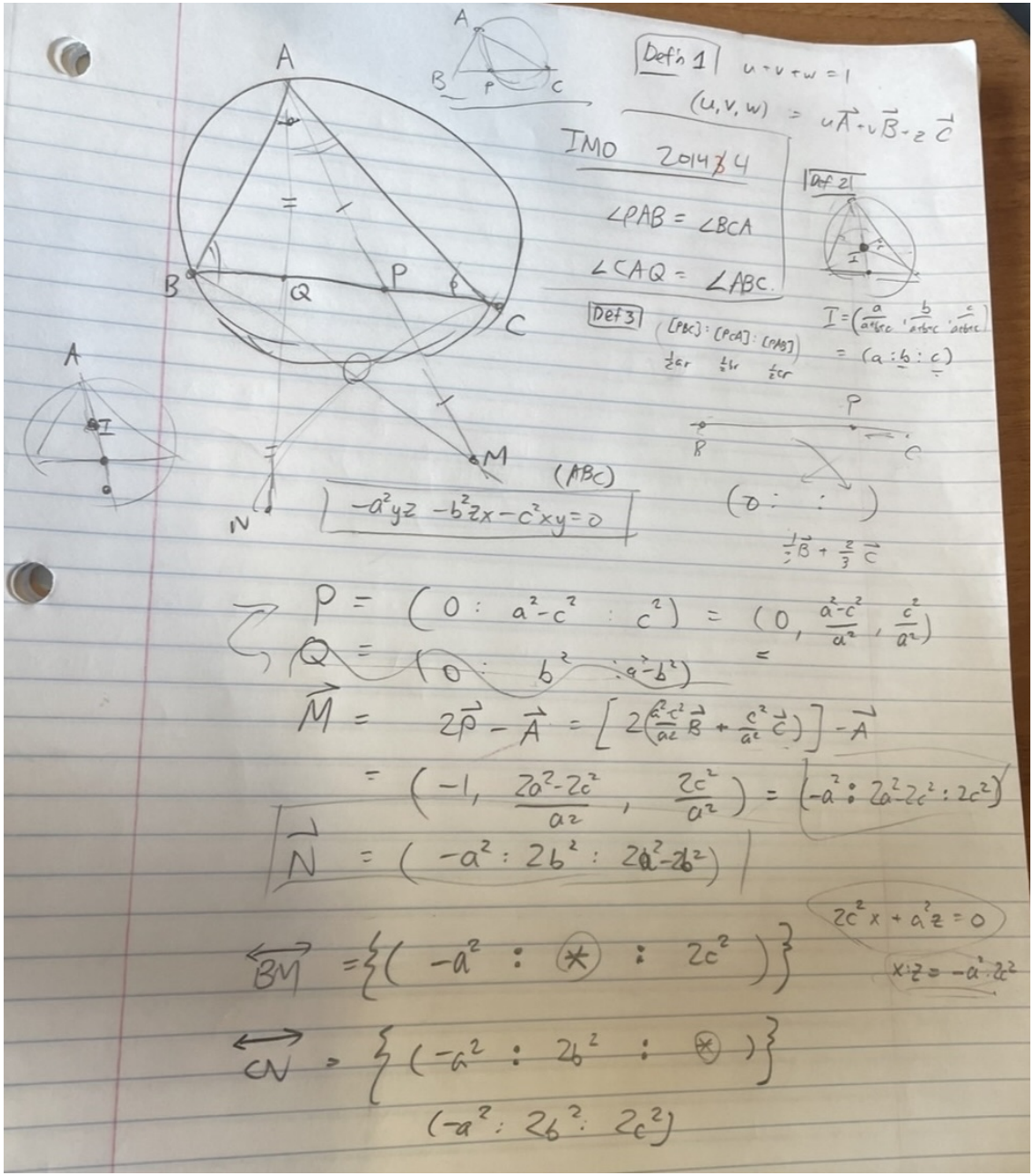
if two elements in same row, same value

or two elements in same column, same value $\rightarrow q$.

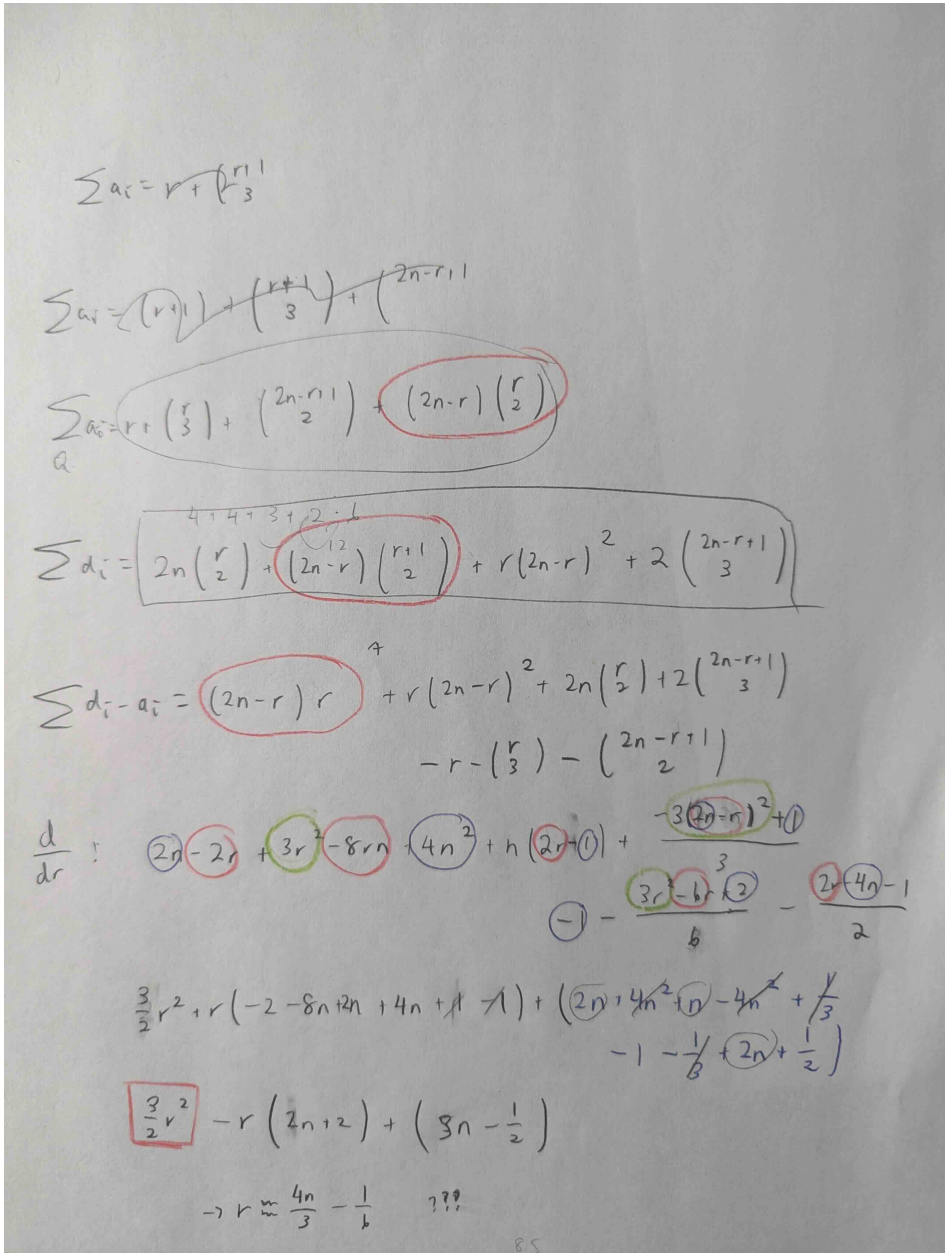
Selena Ge


3, 17, 99, 577, 3363, 19601, 114243, 665857
3880899, 22619537, 131836323
a(n)=6a(n-1)-a(n-2)
4 mod 5
0 mod 3
3 mod 6
3, 2, 4, 2, 3, 1, 3, 2, 4, 2, 3
2^n + 2

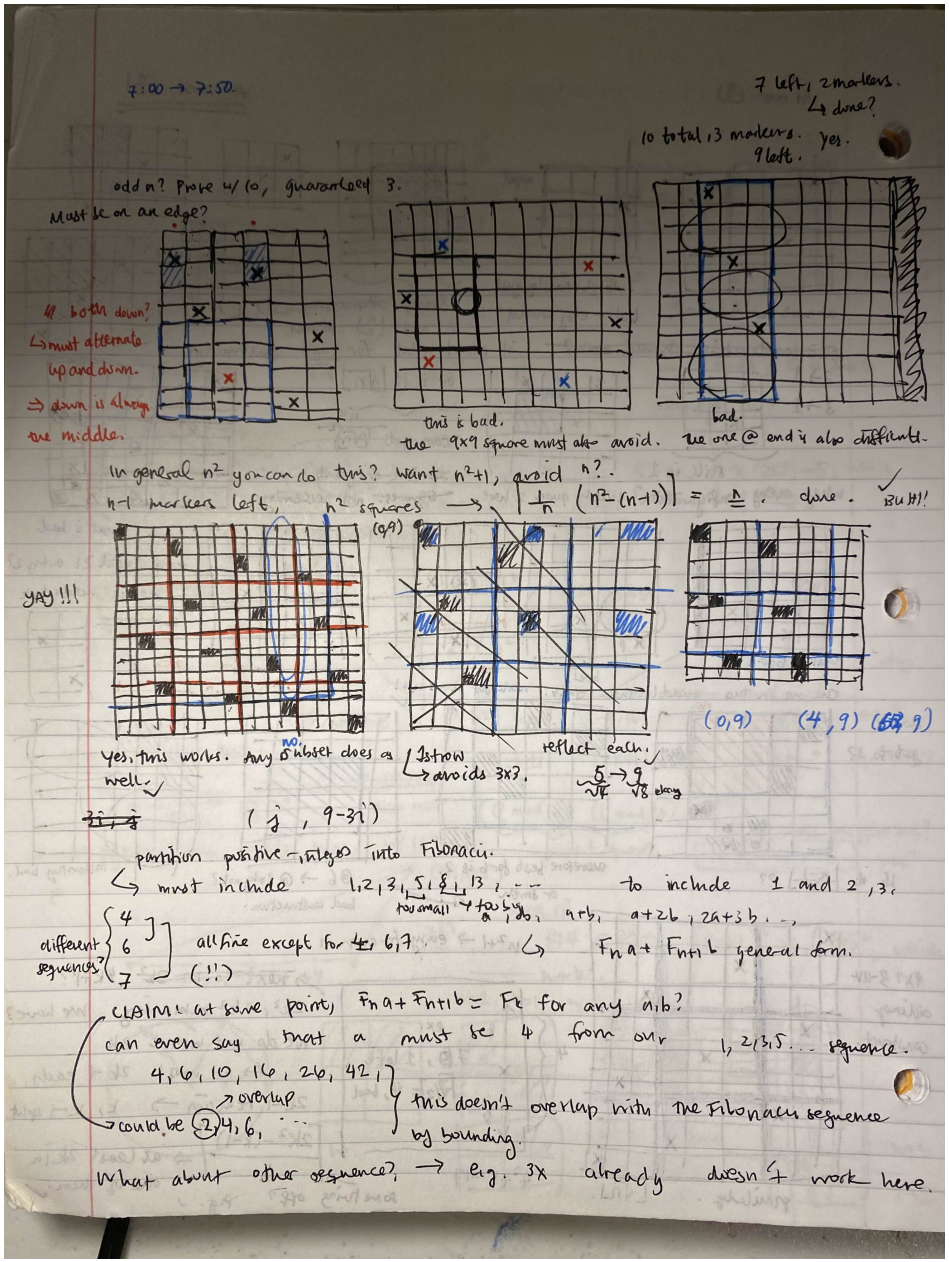
Evan Chen


A
B
C
P
Q
M
N
I
Def 1
u + v + w = 1
(u, v, w) = uA + vB + wC
IMO 2014/4
∠PAB = ∠BCA
∠CAQ = ∠ABC
Def 2
Def 3
[PBC] : [PCA] : [PAB]
(ABC)
−a²yz − b²zx − c²xy = 0
P = (0 : a² − c² : c²) = (0, (a² − c²)/a², c²/a²)
Q = (0 : b² : a² − b²)
M = 2P − A = [2((a² − c²)/a² B + c²/a² C)] − A
= (−1, (2a² − 2c²)/a², 2c²/a²) = (−a² : 2a² − 2c² : 2c²)
N = (−a² : 2b² : 2a² − 2b²)
BM = {(−a² : ⊛ : 2c²)}
CN = {(−a² : 2b² : ⊛)}
(−a² : 2b² : 2c²)
2c²x + a²z = 0
x : z = −a² : 2c²

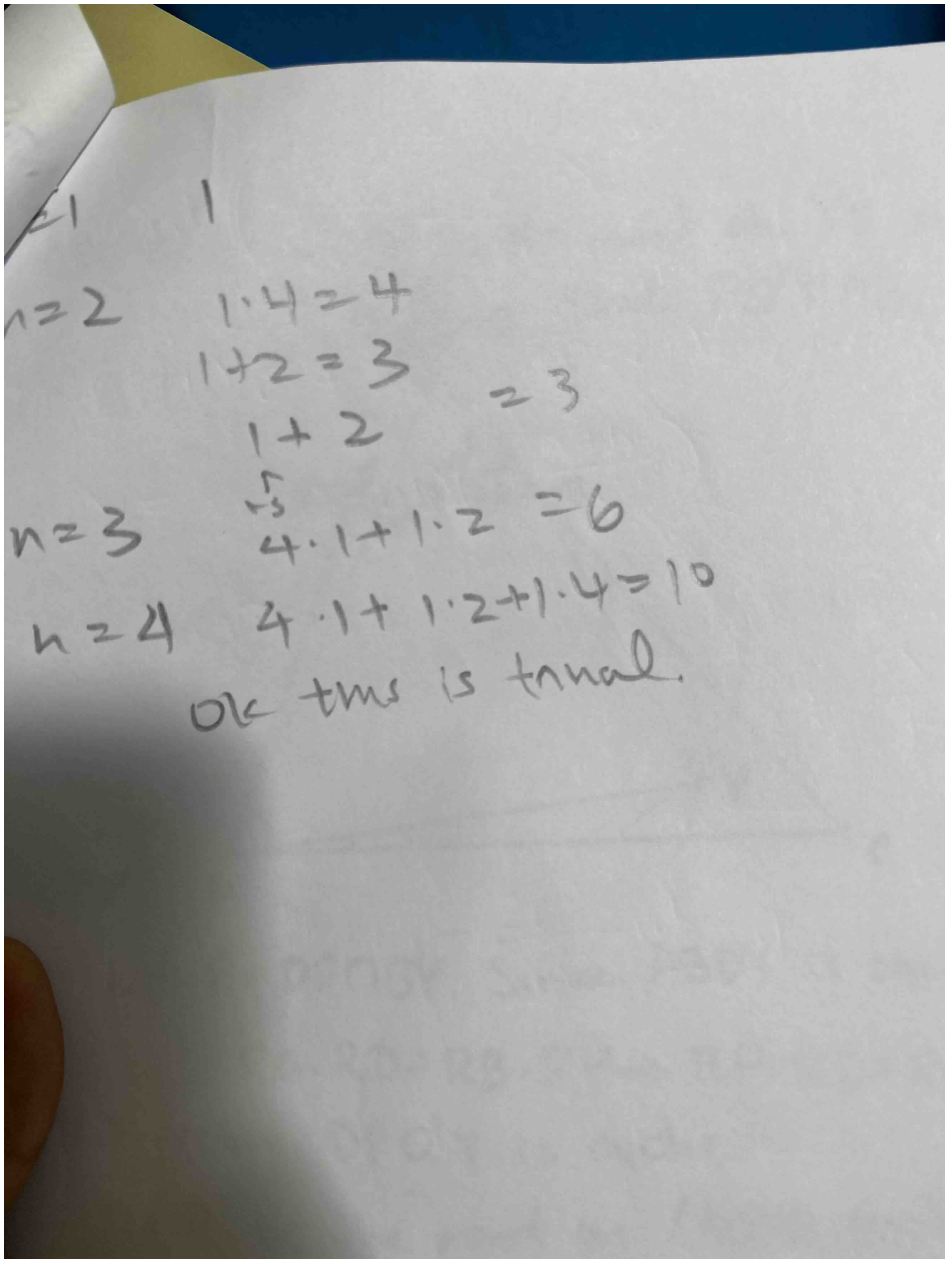
Jiahe Liu


$\sum a_i = r + \binom{r+1}{3}$
$\sum a_i = \binom{r+1}{2} + \binom{r+1}{3} + \binom{2n-r+1}{2}$
$\sum a_i = r + \binom{r}{3} + \binom{2n-r+1}{2} + (2n-r)\binom{r}{2}$
$\sum d_i = 2n\binom{r}{2} + (2n-r)\binom{r+1}{2} + r(2n-r)^2 + 2\binom{2n-r+1}{3}$
$\sum d_i - a_i = (2n-r)r + r(2n-r)^2 + 2n\binom{r}{2} + 2\binom{2n-r+1}{3} - r - \binom{r}{3} - \binom{2n-r+1}{2}$
$\frac{d}{dr}$:
$\frac{3}{2}r^2 + r(-2-8n+2n+4n+1-1) + (2n+4n^2+n-4n+\frac{1}{3} - 1 - \frac{1}{6} + 2n + \frac{1}{2})$
$\frac{3}{2}r^2 - r(2n+2) + (3n - \frac{1}{2})$
$\rightarrow r \approx \frac{4n}{3} - \frac{1}{6}$???

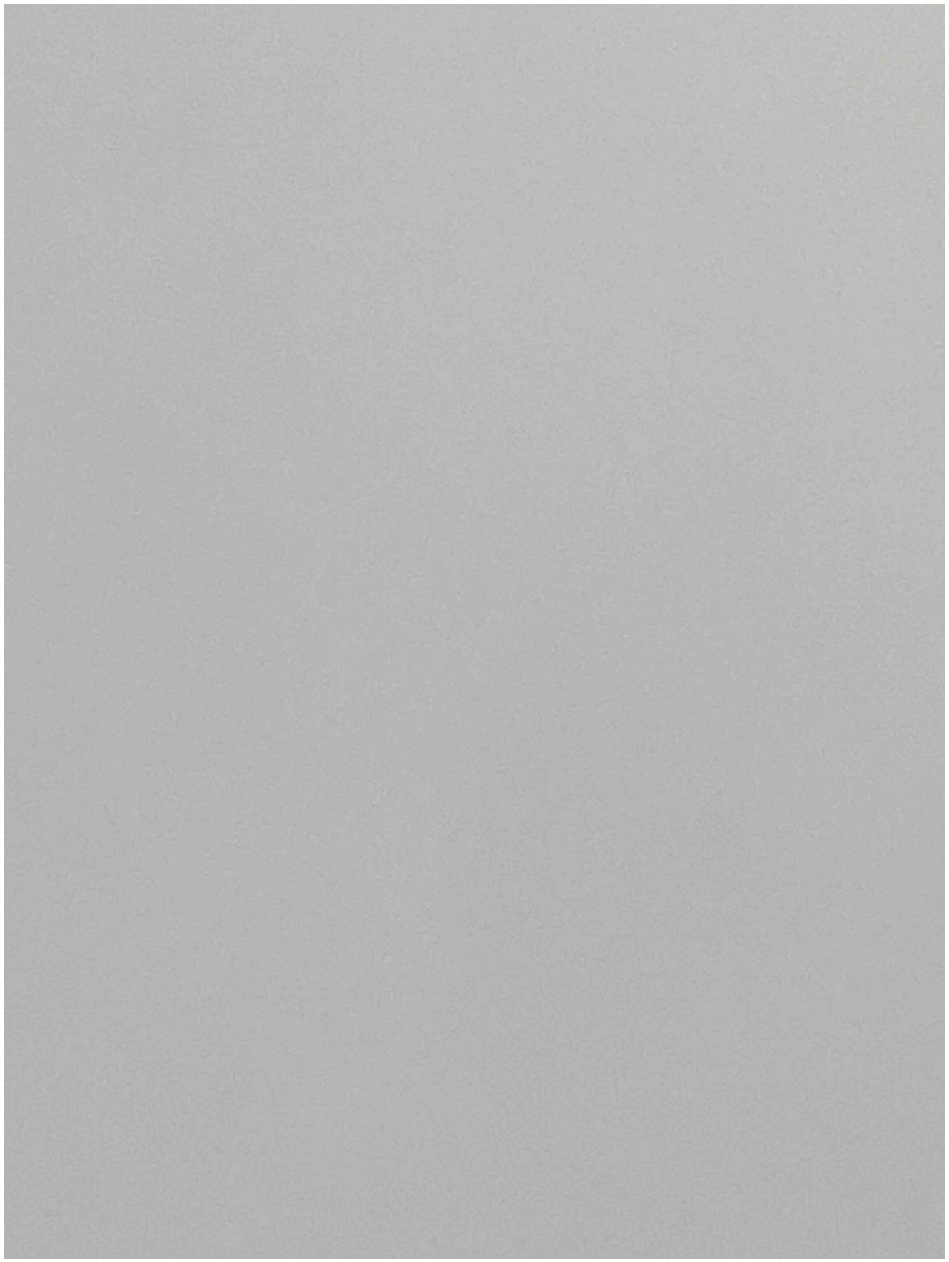
Catherine Xu



Andrew Lin



Alexander Wang



Practice meet today

We'll do Sample Meet 4 Coyote. Write down reasoning in words. I'll check! If you do not do this, you will face execution.

Good luck!